

Causal-MoE: Aligning Reasoning Structure with Computation in LLMs

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Abstract—Large Language Models (LLMs) exhibit strong reasoning capabilities but remain computationally inefficient due to uniform token-level processing, where all tokens incur similar compute regardless of their semantic role, and to address this limitation, we propose our framework Causal-MoE, a framework that replaces token-level routing with *semantic step-wise expert routing*, decomposing outputs into atomic reasoning steps and dynamically assigning each step to a specialised expert, and we implement this design on Qwen2.5-7B-Instruct by replacing selected Feed-Forward Network (FFN) layers with sparse gated MoE modules composed of LoRA-based experts for [MATH], [LOGIC], [COMMONSENSE], and [VERIFY], and incorporate Probability of Necessity and Sufficiency (PNS) to refine routing decisions and enable causal attribution of reasoning steps. We further evaluate two system variants: an Exploded multi-call pipeline for fine-grained control and a Collapsed single-pass formulation for efficient deployment. Experimental results on GSM8K and CommonsenseQA show that Causal-MoE reduces decoding overhead and achieves up to 34% lower estimated FLOPs (2.81T vs. 4.28T) while maintaining competitive accuracy (76.80% on GSM8K) and improving task performance in the routed setting (83.05% on CommonsenseQA), these findings demonstrate that aligning expert routing with reasoning structure enables an effective trade-off between efficiency, interpretability, and performance within a 7B-parameter model.

Index Terms—Mixture-of-Experts (MoE), Causal-MoE, Step-wise Routing, Probability of Necessity and Sufficiency (PNS), and Sparse Routing.

1. Introduction

Autoregressive LLMs generate tokens sequentially while applying nearly uniform computation to each token, regardless of its semantic role. This design is effective for language modelling but inefficient for multi-step reasoning, where computational demand varies significantly across steps, and

we term this mismatch **Compute Reasoning Displacement**: the model expends comparable FLOPs on low-information tokens (e.g., connective phrases) and high-value reasoning operations (e.g., multi-step algebra or logical inference). This inefficiency manifests in two system-level issues: **Latency Explosion**, where Chain-of-Thought (CoT) prompting increases output length and decoding cost (e.g., 208.9 tokens on GSM8K with many low-utility tokens), and **Reasoning Opacity**, where outputs are generated as unstructured sequences without clear sub-task boundaries, hindering attribution, debugging, and targeted intervention. Together, these limitations indicate a misalignment between the computational pathway of current LLMs and the structure of the reasoning tasks they are intended to solve. Existing sparse Mixture-of-Experts (MoE) models rely on token-level routing, which overlooks the semantic structure of multi-step reasoning and often results in inefficient expert utilization and limited interpretability that’s why We propose **Causal-MoE**, a step-wise expert routing framework that operates on *atomic reasoning steps* instead of tokens. Each step is assigned a semantic type (e.g., [MATH], [LOGIC], [VERIFY]) and routed to a specialized expert, enabling structured and interpretable reasoning aligned with task decomposition. Our idea is to divide the reasoning process into specific, explicitly-typed steps, with each step handled by a dedicated expert, by this method builds on concepts from sparse Mixture-of-Experts [1], [3] and structured Chain-of-Thought [4]. What distinguishes our approach is that routing decisions are made based on the semantic meaning of each reasoning step, rather than at the token level. This design reduces redundant generation and improves both efficiency and transparency. To support this, we introduce an **Atomic Reasoning Trace (ART)** format for structured supervision and integrate **Probability of Necessity and Sufficiency (PNS)** to quantify the causal contribution of reasoning steps and expert decisions. We further explore two inference paradigms: an **Exploded (V1)** multi-call setup for fine-grained control and a **Collapsed (V2)** single-pass formulation for efficient deployment. Empirically, our approach

reduces decoding steps by up to **70–80%** and achieves approximately **34% FLOP reduction** on GSM8K while maintaining effective reasoning performance.

1.1. Contributions

- We introduce semantic step-wise expert routing, shifting MoE design from token-level to reasoning-level granularity.
- We propose the Causal-MoE architecture, integrating sparse expert MLPs without modifying the underlying attention backbone.
- We develop the Atomic Reasoning Trace (ART) format for structured supervision and transparent expert attribution.
- We incorporate necessity sufficiency-based causal analysis (PNS) to evaluate the contribution of reasoning steps and expert decisions.
- We design and analyze two inference strategies Exploded (V1) and **Collapsed (V2)**, for different efficiency control trade-offs.
- We demonstrate substantial efficiency gains ($\approx 34\%$ FLOP reduction and significant decoding compression) on GSM8K while maintaining strong reasoning behavior.

2. Related Work

2.1. Sparse Mixture-of-Experts

Sparse Mixture-of-Experts (MoE) architectures such as Switch Transformer, Mixtral-8x7B, and DeepSeek-MoE improve scalability by routing tokens to a subset of experts via a learned gating function [1], [3], [8], [10]. These systems primarily optimize throughput and parameter efficiency through token-level load balancing. However, token-level routing is agnostic to the semantic structure of multi-step reasoning, leading to suboptimal expert utilization for reasoning-intensive tasks. In contrast, we perform routing at the *reasoning-step level*, selecting experts based on the semantic role of each step, which better aligns computation with task structure.

2.2. Chain-of-Thought and Structured Reasoning

Chain-of-Thought (CoT) prompting elicits intermediate reasoning steps and improves accuracy on complex tasks [4], but increases output length and decoding cost. Methods such as self-consistency [17] and Tree-of-Thoughts [18] further enhance reasoning via multi-path exploration, incurring additional compute and latency overhead. Program-Aided Language Models (PAL) [5] introduce structured reasoning with external tool execution. Our approach instead maintains an *internal* structured trace and couples it with expert routing, avoiding external calls while reducing token overhead and improving step-level attribution.

2.3. Parameter-Efficient Fine-Tuning (PEFT)

Parameter-efficient fine-tuning methods, notably LoRA [2], adapt large models by injecting low-rank updates while keeping base weights frozen. We leverage LoRA to instantiate specialized experts within our MoE layers, enabling efficient domain specialization without modifying the attention backbone or incurring full fine-tuning costs.

3. The Causal-MoE Architecture

3.1. Overview and Injection Strategy

We build Causal-MoE on a 4-bit quantized version of Qwen2.5-7B-Instruct to ensure computational feasibility during training and deployment [6], [13] and the base Transformer architecture [9] remains largely unchanged, and modifications are introduced in a targeted manner to preserve pretrained representations. Specifically, we replace the SwiGLU-based Feed-Forward Network (FFN) sublayers at selected depths $\mathcal{L} = \{6, 12, 18, 24\}$ with custom CausalMoEMLP modules. These layers are chosen to balance representational capacity and routing effectiveness, acting as key transformation points within the network. To study routing strategies, we design two system variants:

- **V1 (Exploded)**: A modular multi-model setup where a smaller Qwen2.5-1.5B-Instruct model serves as an external routing controller. It predicts the expert type for each reasoning step, while the main 7B model executes the corresponding computation.
- **V2 (Collapsed)**: A unified architecture in which routing is embedded directly within the 7B model. This eliminates the need for an external controller and enables efficient single-pass inference.

3.2. Expert Design and Routing Mechanism

We implement four specialized experts using LoRA adapters, each corresponding to a distinct reasoning function:

- **MATH**: Numerical computation and algebraic reasoning
- **LOGIC**: Step-wise deduction and rule-based reasoning
- **COMMONSENSE**: Everyday knowledge and contextual inference
- **VERIFY**: Output validation and formatting consistency

Each replaced FFN layer is augmented with a gating network that selects experts dynamically. Given a hidden representation $\mathbf{h}$, the gating function is defined as:

$$G(\mathbf{h}) = \mathbf{h}W_{\text{gate}} \in \mathbb{R}^{N_e} \quad (1)$$

Causal-MoE Sytem Architecture Training Pipeline: Surgery & Causal Distillation

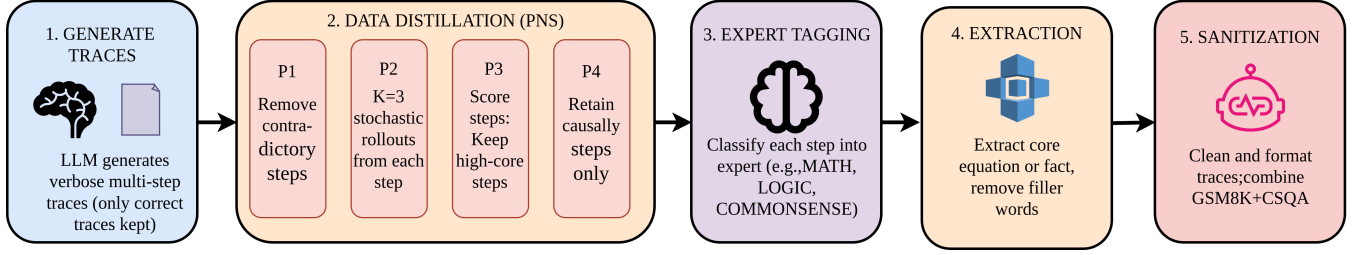


Figure 1. Training pipeline for Causal-MoE. The process begins with trace generation (GEN) using Qwen2.5-7B, followed by a PNS-based data distillation stage that evaluates the necessity of each reasoning step via contradiction rollouts. Redundant steps are removed, and the remaining steps undergo expert tagging (TAG), equation extraction (EXTRACT), and sanitization (SANITIZE) to produce the final training dataset.

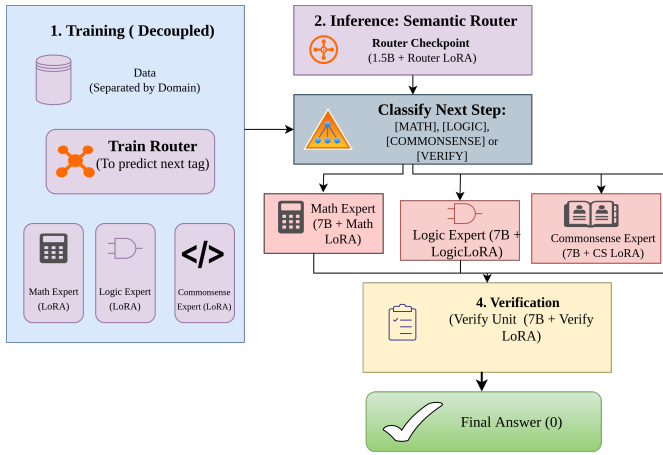


Figure 2. Causal-MoE V1 (Exploded) architecture. A smaller routing model predicts expert labels for each reasoning step, and the main model dynamically loads the corresponding LoRA adapter to generate step-wise outputs.

The top- $k = 2$ experts are selected based on softmax-normalized scores:

$$\mathbf{p} = \text{Softmax}(G(\mathbf{h})), \quad (j_1, j_2) = \text{top-}k(\mathbf{p}) \quad (2)$$

The final output is computed as a weighted combination of selected experts:

$$\text{MoE}(\mathbf{h}) = \sum_{j \in \{j_1, j_2\}} \frac{p_j}{p_{j_1} + p_{j_2}} \cdot \text{Expert}_j(\mathbf{h}) \quad (3)$$

This design preserves the efficiency of sparse MoE while introducing semantic specialization at the reasoning-step level.

3.3. Expert Initialization via Interleaved Weight Slicing

To retain pretrained knowledge while enabling expert specialisation, we initialise experts using *interleaved weight slicing*. The original FFN weights are partitioned into disjoint subsets $S_i \subset \{0, \dots, d_{\text{FFN}} - 1\}$ and assigned to different

experts by ensuring that each expert inherits meaningful pre-trained representations, thereby improving convergence stability and reducing the need for extensive retraining. so this architecture enables a direct alignment between semantic reasoning structure and computational pathways, addressing computational reasoning displacement at the system level.

4. Training Methodology and PNS Scoring

4.1. Atomic Reasoning Trace Format

To enforce structured and efficient reasoning, we introduce the *Atomic Reasoning Trace (ART)* format and each reasoning step is explicitly tagged with a semantic label (e.g., [MATH], [LOGIC]) followed by the corresponding operation (e.g., [MATH] $16 - 3 = 13$), and this design serves two purposes: (i) it enables direct activation of specialized experts during training and inference, and (ii) it eliminates redundant conversational tokens, resulting in compact and information-dense reasoning traces.

4.2. Training Protocol

Training Causal-MoE requires supervision over structured reasoning traces. We construct a dataset of over 100,000 *exploded* reasoning traces derived from GSM8K and CommonsenseQA benchmarks. These traces provide fine-grained supervision for both reasoning and expert routing.

- **Precision:** BF16 training with Paged AdamW 8-bit optimizer [14].
- **Learning Rate:** 5×10^{-6} with linear warmup.
- **Loss Masking:** Cross-entropy loss is computed only over generated response tokens, ensuring focused optimization on reasoning outputs.

4.3. PNS Estimation for Step Importance

To ensure that reasoning traces contain only causally relevant steps, we adopt the **Probability of Necessity and**

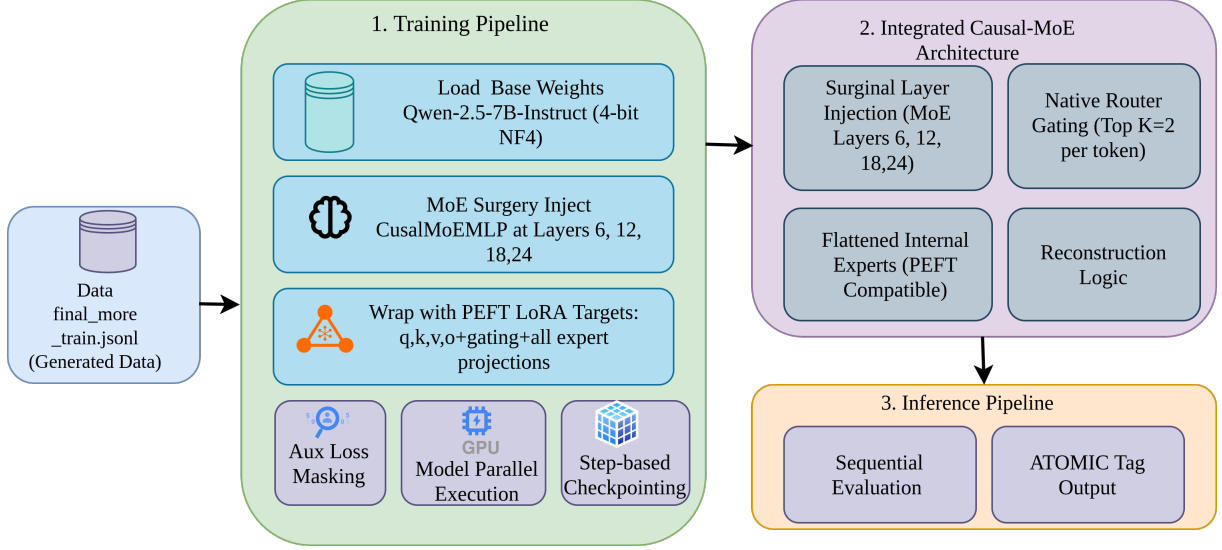


Figure 3. Causal-MoE V2 (Collapsed) architecture. Custom `CausalMoEMLP` modules are integrated at layers 6, 12, 18, and 24, where a gating network dynamically selects the top-2 experts for each input.

TABLE 1. CONFIGURATION FOR LORA AND TRAINING

Hyperparameter	Value
Backbone	Qwen2.5-7B-Instruct (4-bit NF4)
Learning Rate	5×10^{-6} (Linear Warmup)
LoRA Rank/Alpha	32 / 64
Surgical Layers	[6, 12, 18, 24]
Optimizer	Paged AdamW 8-bit

Sufficiency (PNS) framework based on counterfactual causation [16]. Given a reasoning step s_i within a trace that produces the correct answer y^* for a query q , we estimate its importance as:

$$\text{PNS}(s_i) = 1 - \frac{1}{K} \sum_{k=1}^K \mathbf{1}[\hat{y}_k = y^*] \quad (4)$$

where $\hat{y}_k$ denotes the output obtained under a perturbed rollout, and $K = 3$ is the number of counterfactual evaluations. A high PNS score indicates that the step is critical for producing the correct outcome.

4.3.1. Sabotage-Based Rollout Strategy. Instead of removing steps, which allows models to recover via implicit reasoning, we introduce an active perturbation strategy. For each step s_i , we inject an explicit contradiction (e.g., “*ERROR: The previous reasoning step is incorrect*”) and observe whether the model can still recover the correct answer. This approach provides a more reliable estimate of step necessity by enforcing logical inconsistency during evaluation.

4.3.2. Algorithm: CoT Optimization via Necessity and Sufficiency.

Algorithm 1 Optimizing CoT via Necessity and Sufficiency

Require: Initial CoT chain S_{init} , ground truth y , query q , threshold $\alpha = 0.5$

Ensure: Optimized CoT chain S_{final}

```

1:  $\hat{y}_{init} \leftarrow \text{Rollout}(S_{init}, q)$ 
2:  $PS \leftarrow \mathbf{1}[\hat{y}_{init} = y]$ 
3: if  $PS = 1$  then
4:    $S_{current} \leftarrow S_{init}$ 
5:   for each step  $s_t \in S_{current}$  do
6:     Generate sabotage prompt for  $s_t$ 
7:     Compute  $K$  rollout scores  $V_{scores}$ 
8:      $\text{PNS}(s_t) \leftarrow 1 - \frac{1}{K} \sum_{v \in V_{scores}} v$ 
9:     if  $\text{PNS}(s_t) \geq \alpha$  then
10:      Append  $s_t$  to  $S_{final}$ 
11:    end if
12:  end for
13: else
14:    $S_{final} \leftarrow S_{init}$ 
15: end if
16: return  $S_{final}$ 

```

4.3.3. Execution Infrastructure. PNS estimation is performed only on correctly solved traces. The system is implemented using vLLM [15] with a Ray backend across 2 GPUs in BF16 precision. Each batch processes $32 \times 4 \times 3 = 384$ prompts for GSM8K and up to twice that for CommonsenseQA. To reduce inference overhead, `max_tokens` is capped at 80, as only final answers are required for evaluation, yielding a 4–5 $\times$ speedup. Answer extraction follows dataset-specific formats: ##### <number> for GSM8K and ##### <letter> for CommonsenseQA. This training pipeline enforces causally grounded and computation-efficient reasoning by combining structured traces with necessity–sufficiency-based pruning.

5. Inference Paradigms: V1 vs V2

5.1. V1: Exploded Multi-Call Routing

In the V1 setting, inference is performed as an iterative multi-call process in which a routing module predicts the semantic tag for each reasoning step and the base model is repeatedly invoked to generate the corresponding output segment, while this design enforces strong modular separation between experts and allows routing decisions to be conditioned on the full context, it introduces significant computational overhead due to repeated prefill and decoding operations. As a result, latency increases substantially with the number of reasoning steps.

5.2. V2: Collapsed Single-Pass Integration

In contrast, V2 integrates the routing mechanism directly within the model, enabling end-to-end generation in a single forward pass by dynamically routing computation based on semantic tags (e.g., [MATH], [LOGIC]) and activating the appropriate expert weights without requiring external control flow or repeated model invocations. This design eliminates multi-call overhead and significantly reduces inference cost, leading to improved efficiency and lower latency while preserving structured reasoning behavior. The computational cost of inference can be approximated as:

$$\text{FLOP}(\text{model}, T) \approx 2 \times P_{\text{active}} \times T \quad (5)$$

where P_{active} denotes the number of active parameters per token (due to sparse expert selection), and T is the total number of generated tokens. By reducing both P_{active} and T , the collapsed formulation achieves substantial efficiency gains.

6. Experiments & Results

We report both accuracy and system-level efficiency to evaluate practical deployment trade-offs.

6.1. Accuracy Benchmarks

We evaluate Causal-MoE on two standard reasoning benchmarks: CommonsenseQA (CSQA) [12] with 1,221 samples and GSM8K [11] with 1,319 samples. These datasets provide complementary evaluation of commonsense reasoning and multi-step mathematical reasoning.

Results show that V1 improves performance on CSQA, indicating that explicit step-wise routing and modular reasoning benefit commonsense tasks. On GSM8K, V2 achieves a significant recovery over V1 while maintaining a more efficient inference profile, demonstrating a better trade-off between reasoning capability and system efficiency.

TABLE 2. ACCURACY COMPARISON

Benchmark	Model	Samples	Accuracy
CSQA	Base (Qwen2.5-7B)	1221	82.06%
CSQA	Causal-MoE (V1)	1221	83.05%
CSQA	Causal-MoE (V2)	1221	78.38%
GSM8K	Base (Qwen2.5-7B)	1319	88.86%
GSM8K	Causal-MoE (V1)	1319	68.08%
GSM8K	Causal-MoE (V2)	1319	76.80%

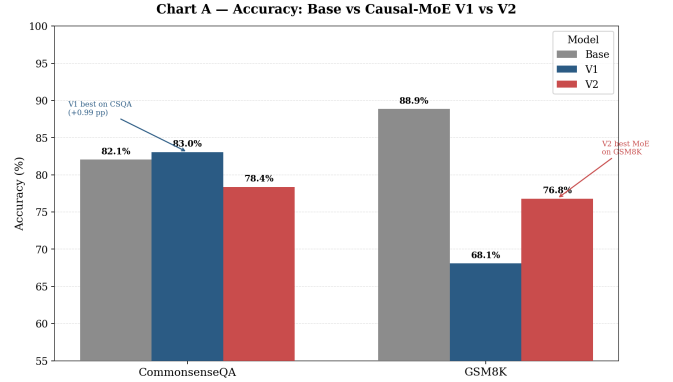


Figure 4. Accuracy comparison across models. V1 improves performance on CSQA, while V2 achieves stronger recovery on GSM8K with improved efficiency.

6.2. Efficiency and Latency Analysis

We evaluate efficiency in terms of decoding steps, generated tokens, and estimated FLOPs. The V2 model significantly reduces computational overhead compared to both the base model and V1.

On GSM8K, V2 reduces estimated FLOPs from 4.28T (base) to 2.81T and representing a $\sim 34\%$ improvement. Compared to V1, the reduction is even more pronounced (7.67T $\rightarrow$ 2.81T), highlighting the cost of multi-call inference. These gains are driven by reductions in both token generation and repeated model invocations.

6.3. Reasoning Efficiency

To quantify efficiency–performance trade-offs, we define **Reasoning Density** as $\rho_R = \frac{\text{Accuracy}}{\text{TFLOPs}}$, under which V2 achieves a score of 27.33 on GSM8K compared to 20.76 for the base model (a 31.7% improvement), indicating that

TABLE 3. LATENCY AND EFFICIENCY METRICS

Metric	CSQA			GSM8K		
	Base	V1	V2	Base	V1	V2
Decoding Steps	155.4	32.3	33.4	208.9	60.3	75.8
Generated Tokens	155.4	27.1	33.4	208.9	55.0	75.8
Avg. Steps	1	5.1	3.9	1	5.3	1.0
Est. FLOPs	3.30T	8.23T	3.09T	4.28T	7.67T	2.81T

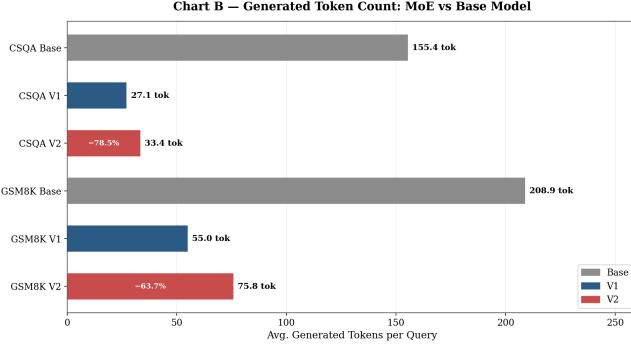


Figure 5. Token generation comparison. V2 reduces token usage by up to 78.5% on CSQA and 63.7% on GSM8K compared to the base model.

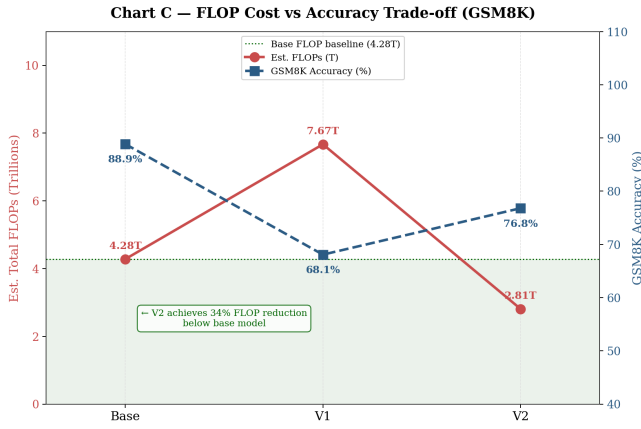


Figure 6. Accuracy vs. FLOP trade-off. V2 achieves a lower computational footprint while maintaining competitive performance, particularly on GSM8K.

Causal-MoE both reduces computational cost and improves the effectiveness of computation per unit FLOP. Causal-MoE demonstrates that step-wise expert routing enables significant efficiency gains while preserving competitive reasoning performance, with V2 providing the most practical deployment trade-off.

7. Case Analysis: Success and Failure

7.1. CommonsenseQA (Success: V1)

Question: “Banks are usually quite secure; what type of door might they use to enhance security?”

Trace:

- 1) [COMMONSENSE] Revolving doors provide controlled entry.
- 2) [COMMONSENSE] They prevent tailgating and unauthorized access.
- 3) [LOGIC] Controlled entry mechanisms improve security.
- 4) [LOGIC] Banks require strict security measures.
- 5) [VERIFY] ##### A (Revolving Door)

Outcome: Correct. The model combines commonsense reasoning with logical inference to produce a compact and coherent reasoning trace, where explicit step-wise routing enables effective task decomposition, leading to accurate and interpretable reasoning.

7.2. GSM8K (Failure Case: V2)

Question: Josh buys a house for \$80,000 and spends \$50,000 on repairs, increasing its value by 150%. What is his profit?

Trace:

- 1) [MATH] $80,000 + 50,000 = 130,000$
- 2) [MATH] $130,000 \times 1.50 = 195,000$
- 3) [MATH] $195,000 - 130,000 = 65,000$
- 4) [VERIFY] ##### 65000

Ground Truth: 70,000.

Analysis: Although the arithmetic operations are executed correctly, the model misinterprets the semantic scope of the percentage increase and specifically we can say, it applies the 150% increase to the total investment rather than the original purchase price, leading to an incorrect final answer, this failure highlights a limitation of the current routing mechanism: while expert specialisation improves local reasoning accuracy, global constraint understanding and semantic interpretation remain challenging. These cases illustrate that step-wise expert routing improves interpretability and modular reasoning, but accurate global reasoning still depends on correctly modelling cross-step semantic dependencies.

8. Discussion

Recent work by Yu et al. [7] applies Probability of Necessity and Sufficiency (PNS) to prune redundant steps in Chain-of-Thought (CoT) reasoning. While effective in reducing token usage (e.g., 78.4% on GSM8K), such approaches operate post-hoc on top of large monolithic models (e.g., DeepSeek-R1 [20]) and do not reduce the underlying computational cost per token. In contrast, Causal-MoE integrates causal reasoning directly into the model architecture by training on PNS-filtered traces and performing semantic step-wise routing, efficiency is enforced during generation rather than applied after inference. This leads to three key advantages: (i) **architectural efficiency**, with reductions in both token usage (63.7% on GSM8K) and per-pass FLOPs (34.3%); (ii) **semantic specialization**, where labeled steps (e.g., [MATH], [LOGIC]) are routed to dedicated experts, improving interpretability; and (iii) **self-contained scalability**, enabling deployment within a single 7B model without reliance on external evaluators.

The effectiveness of Causal-MoE arises from three complementary design choices (i) semantic step-wise routing to

TABLE 4. COMPARISON WITH YU ET AL. [7]

Metric	DeepSeek-R1 (Yu et al.)	Causal-MoE (V2)
Inference Paradigm	Monolithic (Post-Hoc)	Sparse Native Routing
Model Scale	Dense (~671B active)	Sparse (7B active)
CSQA Token Reduction	62.8%	78.5%
GSM8K Token Reduction	78.4%	63.7%
Per-Pass FLOP Reduction	0%	34.3%
Causal Interpretability	Implicit	Explicit Expert Attribution

align computation with reasoning structure, (ii) parameter-efficient fine-tuning (LoRA) for expert specialization without modifying the backbone, and (iii) interleaved weight slicing to preserve pretrained knowledge during initialization—which together enable efficient and stable specialization within a sparse architecture, from a broader perspective, Causal-MoE extends traditional MoE by shifting routing granularity from tokens to reasoning steps, and produces a *compressed CoT* representation that retains essential reasoning while reducing redundancy. This enables a more favorable trade-off between accuracy and computational efficiency. We observe measurable system-level gains: +0.99 accuracy improvement on CommonsenseQA (V1), 34.3% FLOP reduction and 63.7% token reduction on GSM8K (V2), and a 31.7% improvement in reasoning density. Additionally, V2 recovers 8.72 accuracy points over V1 on GSM8K. Beyond these metrics, explicit expert attribution improves interpretability, and the modular design allows extensibility through the addition of new experts without retraining the full model. Despite efficiency gains, Causal-MoE introduces trade-offs in reasoning performance. V2 achieves 76.80% accuracy on GSM8K but does not match the base model (88.86%), indicating that compressing reasoning can lead to loss of fidelity. Furthermore, V2 underperforms V1 on CommonsenseQA, suggesting that single-pass inference reduces the model’s ability to explore alternative reasoning paths. The model also occasionally fails to capture global constraints across multi-step reasoning, leading to semantically incorrect conclusions.

9. Conclusion

We present **Causal-MoE**, a framework that introduces semantic step-wise expert routing for efficient and interpretable LLM reasoning. By integrating sparse expert MLPs and leveraging PNS-based training, the proposed approach reduces inference cost by up to **34%** and improves reasoning density by **31.7%** on GSM8K. These results demonstrate that aligning computation with the structure of reasoning tasks enables a favorable trade-off between efficiency and performance. More broadly, this work highlights the potential of embedding causal and semantic structure directly into model architectures as a pathway toward scalable and transparent reasoning systems by this our proposed framework semantic step-wise routing provides a principled mechanism to bridge the gap between reasoning structure and computational efficiency in large language models. Looking

forward, we identify three key directions for improving the framework. First, **dynamic expert weighting** can enable soft routing and better handling of ambiguous or cross-domain reasoning tasks, second, **self-correction mechanisms** (e.g., Verify+) can be incorporated to detect and correct intermediate reasoning errors during inference and third, **global context integration** can improve consistency across long reasoning chains by enforcing global constraints. Embedding causal reasoning and semantic structure directly into the model architecture offers a more effective and scalable alternative to post-hoc optimization for achieving both efficiency and interpretability in LLM reasoning systems.

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